Biology for Engineers – 4th sem

Module – 1

1. Discuss the various components of Eukaryotic cells.
2. Identify the applications of stem cells.
3. Explain the functions of vitamins.
4. Compare Prokaryotic and Eukaryotic cells.
5. Explain the properties of Carbohydrates.
6. Explain the functions of Lipids.

Module – 2

1. Highlighting the properties of cellulose, justify cellulose as an effective water filter.
2. Explain the working and development of DNA vaccines by taking suitable example.
3. What are Bioplastics? Justify the use of PHA as Bioplastic mentioning its properties and applications.
4. Discuss Meat analogs of protein.
5. Discuss Lipids as cleaning agents.

Module – 3

1. What is Electro Encephalogram (EEG)? Discuss the types of Brain activity detected with EEG. Write any three applications.
2. What are Pace Makers? Explain basic design and construction of Pace Makers.
3. Justify Lungs as purification system.
4. Explain architecture of Rod and Cone cells with suitable diagram.

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Module – 4

1. What is ultrasonography? Explain the uses and working principle.
2. What is lotus leaf effect? Explain the mechanism and applications of super Hydrophobic effect.
3. The structure and design of Kingfisher beak lead to the design of Bullet trains. Explain.
4. Explain the working and applications of Bionic Leaf Technology.

Module – 5

1. Explain the use of Electrical tongue in food science.
2. Explain the advantages and limitations of Artificial Intelligence for disease diagnosis.
3. Explain Bioengineering solutions for muscular dystrophy and Osteoporosis.
4. Explain most commonly used Bioprinting Techniques.

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